

2.3 Trig Sub. Worksheet

1. Find $\int \frac{1}{\sqrt{2+x^2}} dx$

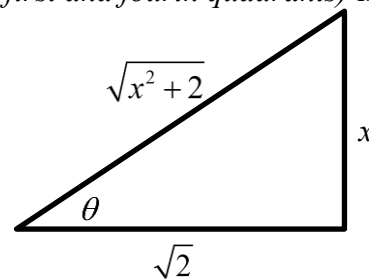
Let $x = \sqrt{2} \tan(\theta)$ which implies that $dx = \sqrt{2} \sec^2(\theta) d\theta$. We restrict θ so that $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. Then,

$$\begin{aligned} \int \frac{1}{\sqrt{2+x^2}} dx &= \int \frac{1}{\sqrt{2+2\tan^2(\theta)}} \sqrt{2} \sec^2(\theta) d\theta = \int \frac{\sec^2(\theta)}{\sqrt{1+\tan^2(\theta)}} d\theta = \int \frac{\sec^2(\theta)}{\sec(\theta)} d\theta \\ &= \int \sec(\theta) d\theta = \ln|\sec(\theta) + \tan(\theta)| + C \end{aligned}$$

We must now write our solution in terms of x . Since $x = \sqrt{2} \tan(\theta) \rightarrow \tan(\theta) = \frac{x}{\sqrt{2}}$ and we can construct the following reference triangle for θ (Note: this triangle is the same in the first and fourth quadrants) is:

From the reference triangle we see that $\sec(\theta) = \frac{\sqrt{x^2+2}}{\sqrt{2}}$ and therefore

$$\ln|\sec(\theta) + \tan(\theta)| + C = \ln\left|\frac{\sqrt{x^2+2}}{\sqrt{2}} + \frac{x}{\sqrt{2}}\right| + C$$



2. Find $\int \frac{1}{x^4 \sqrt{9-x^2}} dx$

Let $x = 3 \sin(\theta)$ which implies that $dx = 3 \cos(\theta) d\theta$. We restrict θ so that $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. Then we have,

$$\begin{aligned} \int \frac{1}{x^4 \sqrt{9-x^2}} dx &= \int \frac{1}{81 \sin^4(\theta) \sqrt{9-9 \sin^2(\theta)}} 3 \cos(\theta) d\theta = \int \frac{\cos(\theta)}{81 \sin^4(\theta) \sqrt{1-\sin^2(\theta)}} d\theta \\ &= \int \frac{1}{81 \sin^4(\theta)} d\theta = \frac{1}{81} \int \csc^4(\theta) d\theta = \frac{1}{81} \int \csc^2(\theta) \csc^2(\theta) d\theta = \frac{1}{81} \int [\cot^2(\theta) + 1] \csc^2(\theta) d\theta \end{aligned}$$

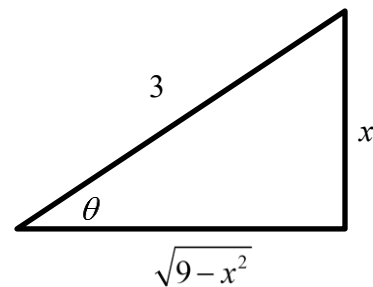
Let $u = \cot(\theta)$ & then $du = -\csc^2(\theta) d\theta \rightarrow -du = \csc^2(\theta) d\theta$

$$\frac{1}{81} \int [\cot^2(\theta) + 1] \csc^2(\theta) d\theta = \frac{1}{81} \int -[u^2 + 1] du = -\frac{1}{81} \left[\frac{1}{3} u^3 + u \right] + C = -\frac{1}{81} \left[\frac{1}{3} \cot^3(\theta) + \cot(\theta) \right] + C$$

We must now write our solution in terms of x . Since $x = 3\sin(\theta) \rightarrow \sin(\theta) = \frac{x}{3}$ and we can construct the following reference triangle for θ (Note: this triangle is the same in the first and fourth quadrants) is:

From the reference triangle we see that $\cot(\theta) = \frac{\sqrt{9-x^2}}{x}$ and therefore

$$-\frac{1}{81} \left[\frac{1}{3} \cot^3(\theta) + \cot(\theta) \right] + C = -\frac{1}{81} \left[\frac{1}{3} \left(\frac{\sqrt{9-x^2}}{x} \right)^3 + \frac{\sqrt{9-x^2}}{x} \right] + C$$



3. Find $\int \frac{\sqrt{x^2-9}}{x} dx$ (assume $x > 0$)

We will proceed with a trig. substitution. Let $x = 3\sec(\theta)$ which implies that $dx = 3\sec(\theta)\tan(\theta)d\theta$.

Remember that $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$, so we cannot solve for θ since the range of the inverse secant does not match these ranges.

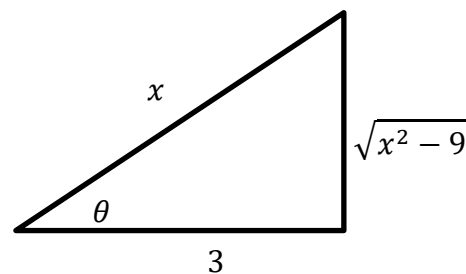
$$\begin{aligned} \int \frac{\sqrt{x^2-9}}{x} dx &= \int \frac{\sqrt{9\sec^2(\theta)-9}}{3\sec(\theta)} 3\sec(\theta)\tan(\theta)d\theta = \int 3\sqrt{\sec^2(\theta)-1}\tan(\theta)d\theta \\ &= \int 3\tan^2(\theta)dx = 3\int [\sec^2(\theta)-1]d\theta = 3[\tan(\theta)-\theta] + C = 3\left[\frac{\sqrt{x^2-9}}{3} - \sec^{-1}\left(\frac{x}{3}\right)\right] + C \end{aligned}$$

Note: Since we assumed $x > 0$ we can be sure that $0 \leq \theta < \frac{\pi}{2}$ and therefore

we guarantee our triangles in the first quadrant and thus $\theta = \sec^{-1}\left(\frac{x}{3}\right)$.

We could also express θ in many other ways here, such as:

$$\theta = \tan^{-1}\left(\frac{\sqrt{x^2-9}}{3}\right) \text{ or } \theta = \sin^{-1}\left(\frac{\sqrt{x^2-9}}{x}\right)$$



4. Find $\int_0^1 \sqrt{1-x^2} dx$.

To evaluate this integral we will use a trig. sub. Let $x = \sin(\theta)$ where $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$, then $dx = \cos(\theta)d\theta$.

$$0 = \sin(\theta) \Leftrightarrow \theta = 0$$

Our bounds become:

$$1 = \sin(\theta) \Leftrightarrow \theta = \frac{\pi}{2} \text{ . Therefore we have:}$$

$$\begin{aligned} \int_0^1 \sqrt{1-x^2} dx &= \int_0^{\frac{\pi}{2}} \sqrt{1-\sin^2(\theta)} \cdot \cos(\theta) d\theta = \int_0^{\frac{\pi}{2}} \cos^2(\theta) d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} [1 + \cos(2\theta)] d\theta = \frac{1}{2} \left[\theta + \frac{1}{2} \sin(2\theta) \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left[\frac{\pi}{2} + \frac{1}{2} \sin(\pi) - 0 - \frac{1}{2} \sin(0) \right] = \frac{\pi}{4} \end{aligned}$$

5. Find $\int \frac{dt}{\sqrt{t^2-6t+13}}$.

In order to make this integral doable with trigonometric substitution, we complete the square as follows:

$$t^2 - 6t + 13 = (t^2 - 6t + 9) - 9 + 13 = (t-3)^2 + 4 \text{ (Note: you should recall that in order to complete the square for the expression } x^2 + bx \text{ you must add } \left(\frac{b}{2}\right)^2 \text{ to the expression).}$$

We can now rewrite the integral as: $\int \frac{dt}{\sqrt{t^2-6t+13}} = \int \frac{dt}{\sqrt{(t-3)^2+4}}$. From here we make the substitution:

$$u = t-3 \Leftrightarrow du = dt \text{ , and we obtain: } \int \frac{dt}{\sqrt{(t-3)^2+4}} = \int \frac{du}{\sqrt{u^2+4}} \text{ .}$$

The integral in terms of u is now set up for trig substitution. To evaluate this integral we let $u = 2 \tan(\theta)$ where $-\frac{\pi}{2} < \theta < \frac{\pi}{2}$. This implies $\theta = \tan^{-1}\left(\frac{u}{2}\right)$ and $du = 2 \sec^2(\theta) d\theta$.

$$\text{Now, with this substitution we have: } \frac{1}{\sqrt{u^2+4}} = \frac{1}{\sqrt{4 \tan^2(\theta)+4}} = \frac{1}{2 \sec(\theta)} \text{ .}$$

Now, we find the integral:

$$\int \frac{du}{\sqrt{u^2+4}} = \int \frac{2 \sec^2(\theta)}{2 \sec(\theta)} d\theta = \int \sec(\theta) d\theta = \ln|\sec(\theta) + \tan(\theta)| + C \text{ .}$$

We must now write our solution in terms of t . To do this we first write our answer in terms of u . Since

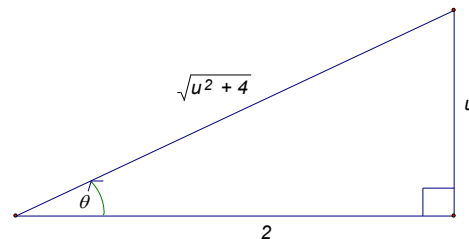
$\tan(\theta) = \frac{u}{2}$, the reference triangle for θ (Note: this triangle is the same in the first and fourth quadrants) is:

From the reference triangle we see that:

$$\begin{aligned}\ln|\sec(\theta) + \tan(\theta)| + C &= \ln\left|\frac{\sqrt{u^2+4}}{2} + \frac{u}{2}\right| + C = \ln\left|\frac{1}{2}(\sqrt{u^2+4} + u)\right| + C \\ &= \ln\left|(\sqrt{u^2+4} + u)\right| + \ln\left|\frac{1}{2}\right| + C = \ln|\sqrt{u^2+4} + u| + C'.\end{aligned}$$

To finish the problem we use $u = t - 3$ to get our answer in terms of t :

$$\begin{aligned}\ln|\sqrt{u^2+4} + u| + C' &= \ln|\sqrt{(t-3)^2+4} + t-3| + C'. \text{ Thus, the} \\ \text{answer is: } \ln|\sqrt{(t-3)^2+4} + t-3| + C'.\end{aligned}$$



6. Find the exact value of $\int_{-6}^{-3\sqrt{2}} \frac{\sqrt{x^2-9}}{x} dx$

We will proceed with a trig. substitution. Let $x = 3\sec(\theta)$ which implies $dx = 3\sec(\theta)\tan(\theta)d\theta$.

Remember that $0 \leq \theta < \frac{\pi}{2}$ or $\pi \leq \theta < \frac{3\pi}{2}$, so we cannot solve for θ since the range of the inverse secant does not match these ranges.

Notice that our bounds change: $\sec(\theta) = -\frac{6}{3} = -2 \Rightarrow \theta = \frac{4\pi}{3}$ & $\sec(\theta) = -\frac{3\sqrt{2}}{3} = -\sqrt{2} \Rightarrow \theta = \frac{5\pi}{4}$.

(Remember again we cannot solve for θ since the range inverse secant does not match the given ranges.) Therefore we have:

$$\begin{aligned}\int_{-6}^{-3\sqrt{2}} \frac{\sqrt{x^2-9}}{x} dx &= \int_{\frac{4\pi}{3}}^{\frac{5\pi}{4}} \frac{\sqrt{9\sec^2(\theta)-9}}{3\sec(\theta)} \cdot 3\sec(\theta)\tan(\theta)d\theta = \int_{\frac{4\pi}{3}}^{\frac{5\pi}{4}} 3\sqrt{\tan^2(\theta)} \cdot \tan(\theta)d\theta \\ &= 3 \int_{\frac{4\pi}{3}}^{\frac{5\pi}{4}} \tan^2(\theta)d\theta = 3 \int_{\frac{4\pi}{3}}^{\frac{5\pi}{4}} [\sec^2(\theta)-1]d\theta = 3[\tan(\theta)-\theta]_{\frac{4\pi}{3}}^{\frac{5\pi}{4}} \\ &= 3\left[\tan\left(\frac{5\pi}{4}\right) - \frac{5\pi}{4} - \left(\tan\left(\frac{4\pi}{3}\right) - \frac{4\pi}{3}\right)\right] = 3\left[1 - \frac{5\pi}{4} - \sqrt{3} + \frac{4\pi}{3}\right] = 3 - 3\sqrt{3} + \frac{\pi}{4}\end{aligned}$$